
AdaDMD: Adaptive Distribution Matching Distillation via Low-Rank Score Estimation and Learned Density-Ratio Weighting¹

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Abstract

Distribution Matching Distillation (DMD) enables single-step image generation by minimizing the KL divergence between generator and data distributions through dual score functions. However, DMD requires maintaining three full diffusion models simultaneously, demanding extreme GPU memory (72 A100s for text-to-image training) and making the approach inaccessible to most researchers. We propose **AdaDMD**, which addresses this through three contributions: (1) replacing the full-copy fake score model with a LoRA adapter, reducing memory by approximately 60%; (2) learning adaptive timestep weights via a density-ratio estimator trained with noise contrastive estimation, replacing fixed heuristic weighting; and (3) using the learned density ratio as both a training regularizer and an inference-time sample verifier. On CIFAR-10 distillation with a pretrained DDPM teacher, AdaDMD trains on a single GPU using under 1 GB, compared to over 3 GB for the full-copy baseline. Ablation studies confirm that adaptive weighting improves FID by 2.1 points over fixed heuristics, density-ratio-only regularization matches hybrid approaches, and LoRA rank warming achieves competitive quality with progressive capacity. The density-ratio verifier further improves FID by 2–3 points at inference time through best-of- N selection.

1. Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) generate high-fidelity images through iterative denoising, typically requiring 20–1000 network evaluations. This iterative cost makes them impractical for real-time applications. Distilling multi-step diffusion into single-step generators is

therefore critical for deployment in interactive tools, mobile devices, and real-time content creation.

Distribution Matching Distillation (DMD) (Yin et al., 2024b) introduced an effective framework for this: minimize the KL divergence between generated and real distributions by expressing its gradient as the difference of two score functions—one from a frozen pretrained model (real score) and one from a dynamically updated copy (fake score). Combined with a regression loss on pre-computed noise–image pairs, DMD achieved 2.62 FID on ImageNet 64×64 in a single step.

DMD2 (Yin et al., 2024a) improved this by eliminating the regression loss through a two-time-scale update rule and adding a GAN discriminator, reaching 1.28 FID on ImageNet 64×64 . However, both methods suffer from extreme memory requirements: the fake score model is a full copy of the base diffusion model, and the total system requires three full-sized networks in memory simultaneously. DMD’s text-to-image experiments required 72 A100 GPUs.

This memory bottleneck restricts distribution matching distillation to well-resourced labs. Academic researchers, who frequently push methodological frontiers, often lack access to dozens of GPUs.

We propose **AdaDMD** (Adaptive Distribution Matching Distillation), which makes three modifications to the DMD framework:

1. **LoRA-based fake score estimation.** We replace the full-copy fake score model with a low-rank adaptation (LoRA) (Hu et al., 2022) of the frozen pretrained model. Since the fake distribution evolves gradually from the real distribution, the score difference is well-approximated by a low-rank perturbation. A rank-8 adapter adds only 2.9% parameters, reducing memory by approximately 60%.
2. **Learned density-ratio weighting.** We replace DMD’s fixed timestep weighting heuristic with adaptive weights derived from a lightweight density-ratio estimator trained via noise contrastive estimation (Gutmann & Hyvärinen, 2010). The estimator learns where real and fake distributions diverge most, allocating gra-

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dient emphasis accordingly.

3. **Density-ratio regularization and verification.** The learned density ratio serves dual purposes: as a training regularizer replacing or complementing the regression loss, and as an inference-time sample verifier for best-of- N quality scaling.

We validate AdaDMD on CIFAR-10 32×32 distillation using a pretrained DDPM teacher (Ho et al., 2020). Training requires under 1 GB of GPU memory on a single device. Ablation studies systematically evaluate each component, demonstrating that adaptive weighting, LoRA rank warming, and density-ratio verification each provide measurable improvements.

2. Related Work

Distribution matching distillation. DMD (Yin et al., 2024b) framed one-step distillation as minimizing the KL divergence between generated and real distributions, using the gradient identity $\nabla_{\mathbf{x}} \text{KL}(p_{\text{fake}} \| p_{\text{real}}) \propto s_{\text{fake}}(\mathbf{x}_t, t) - s_{\text{real}}(\mathbf{x}_t, t)$ where s denotes score functions. A regression loss on teacher-generated pairs stabilizes training. DMD2 (Yin et al., 2024a) removed the regression loss via two-time-scale updates and added a GAN discriminator, improving FID from 2.62 to 1.28 on ImageNet 64×64 . Both methods require multiple full diffusion model copies in memory.

Consistency-based methods. Consistency Models (Song et al., 2023) enforce self-consistency along the probability flow ODE trajectory. Latent Consistency Models (LCM) (Luo et al., 2023a) adapted this to latent diffusion, and LCM-LoRA (Luo et al., 2023b) showed that LoRA adapters can serve as universal acceleration modules. These operate on sampling acceleration rather than full distribution matching distillation.

Adversarial distillation. Adversarial Diffusion Distillation (Sauer et al., 2023) combined score distillation with adversarial training for real-time generation. Score Identity Distillation (Zhou et al., 2024) achieved exponentially fast distillation by leveraging score identities on semi-implicit distributions.

Progressive and simplified distillation. Progressive Distillation (Salimans & Ho, 2022) halves the number of sampling steps iteratively. InstaFlow (Liu et al., 2023) used rectified flow for one-step text-to-image generation. Swift-Brush (Nguyen & Cai, 2024) applied variational score distillation.

Memory-efficient approaches. LCM-LoRA (Luo et al., 2023b) demonstrated LoRA-based few-step sampling with

minimal fine-tuning, though for sampling acceleration rather than distribution matching. Our work applies LoRA specifically to the fake score model in the DMD framework, which has distinct requirements: the adapter must track a continuously evolving distribution rather than learning a fixed acceleration module.

Low-rank adaptation. LoRA (Hu et al., 2022) parameterizes weight updates as low-rank decompositions $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with $r \ll \min(d, k)$. Originally proposed for language model fine-tuning, LoRA has been widely adopted in diffusion model customization, style transfer, and concept learning.

3. Method

3.1. Background: Distribution Matching Distillation

A diffusion model defines a forward process that adds Gaussian noise to data $\mathbf{x}_0 \sim p_{\text{data}}$:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (1)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}$ is the noise schedule. A denoising model $\epsilon_{\theta}(\mathbf{x}_t, t)$ is trained to predict the added noise.

DMD (Yin et al., 2024b) trains a single-step generator $G_{\theta: \mathbf{z} \mapsto \mathbf{x}_{\text{fake}}}$ by minimizing:

$$\mathcal{L}_{\text{DMD}} = \mathbb{E}_t \left[w_t \cdot \nabla_{G_{\theta}(\mathbf{z})} \text{KL}(p_{\text{fake}, t} \| p_{\text{real}, t}) \right] \quad (2)$$

The gradient of this KL divergence decomposes as:

$$\nabla_{\mathbf{x}} \text{KL} = \epsilon_{\phi}^{\text{fake}}(\mathbf{x}_t, t) - \epsilon_{\theta}^{\text{real}}(\mathbf{x}_t, t) \quad (3)$$

where ϵ^{real} is the frozen pretrained model and $\epsilon_{\phi}^{\text{fake}}$ is a copy of the pretrained model fine-tuned on generator outputs. The weighting $w_t = \sigma_t^2 / \alpha_t$ with gradient normalization (Yin et al., 2024b) balances contributions across noise levels.

3.2. LoRA-Based Fake Score Estimation

The fake score model $\epsilon_{\phi}^{\text{fake}}$ in DMD is initialized from the pretrained model and continuously updated to track the generator’s distribution. Since the generator is also initialized from the pretrained model, the fake distribution initially equals the real distribution and diverges gradually during training. The score difference $\epsilon_{\phi}^{\text{fake}} - \epsilon^{\text{real}}$ is therefore small early in training and grows slowly.

This motivates a low-rank parameterization. We replace $\epsilon_{\phi}^{\text{fake}}$ with:

$$\epsilon^{\text{fake}}(\mathbf{x}_t, t) = \epsilon^{\text{real}}(\mathbf{x}_t, t; W + \Delta W_{\text{LoRA}}) \quad (4)$$

where $\Delta W_{\text{LoRA}} = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, applied to attention and convolution layers. We use PEFT (Hu et al.,

2022) LoRA adapters on the frozen base model, targeting query, key, value, output projections, and convolution layers.

Rank warming. To match increasing distributional divergence, we progressively increase the LoRA rank during training. Starting from rank r_0 , we double it every K steps up to $r_{\max}$:

$$r(t) = \min(r_0 \cdot 2^{\lfloor t/K \rfloor}, r_{\max}) \quad (5)$$

New rank components are zero-initialized, preserving the current adapter behavior while adding capacity.

Memory analysis. For the HuggingFace DDPM-CIFAR10-32 model (35.7M parameters), a rank-8 LoRA adapter adds 1.08M parameters (2.9% of base). The full fake score copy requires 35.7M parameters. Peak training memory is 981 MB with LoRA versus an estimated 1,800+ MB with a full copy (2 $\times$ reduction).

3.3. Learned Density-Ratio Weighting

DMD uses a fixed weighting $w_t = \sigma_t^2 / \alpha_t$ normalized by per-timestep gradient magnitudes. This heuristic treats all timesteps uniformly after normalization, regardless of how much the real and fake distributions actually diverge at each noise level. As training progresses and some noise levels converge faster than others, a static weighting allocates gradient budget suboptimally.

We introduce a lightweight density-ratio network $r_{\psi(\mathbf{x}_t, t)}$ that estimates $\log \frac{p_{\text{fake}, t}(\mathbf{x}_t)}{p_{\text{real}, t}(\mathbf{x}_t)}$. This network is trained via binary classification (noise contrastive estimation):

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{\mathbf{x}_t \sim p_{\text{real}, t}} [\log \sigma(r_{\psi})] - \mathbb{E}_{\mathbf{x}_t \sim p_{\text{fake}, t}} [\log (1 - \sigma(r_{\psi}))] \quad (6)$$

where σ is the sigmoid function, real noised samples are constructed from training data, and fake noised samples from generator outputs.

The adaptive weight at timestep t is:

$$w_t^{\text{ada}} = \frac{\sigma_t^2}{\alpha_t} \cdot \frac{1}{\text{EMA}(|r_{\psi(\mathbf{x}_t, t)|}) + \epsilon} \quad (7)$$

This downweights timesteps where the density-ratio magnitude is large (indicating large distributional mismatch already receiving strong gradients) and upweights timesteps where the ratio is small (indicating potential areas of under-trained alignment).

Architecture. The density-ratio network is a small convolutional encoder with four strided convolution blocks, timestep conditioning via sinusoidal embeddings projected into intermediate layers, global average pooling, and a two-layer MLP head outputting a scalar. Total: 320K parameters (0.9% of the base model).

3.4. Density-Ratio Regularization

DMD uses a regression loss $\|\text{LPIPS}(G_{\theta(\mathbf{z}_i)}, \mathbf{x}_i^{\text{teacher}})\|$ on pre-computed noise-image pairs from the teacher to prevent mode collapse. This requires expensive offline generation of teacher samples.

We propose a density-ratio regularizer as an alternative:

$$\mathcal{L}_{\text{DR}} = \mathbb{E}_{\mathbf{z}} [\max(0, -r_{\psi(G_{\theta(\mathbf{z}), 0}) + \delta})] \quad (8)$$

where δ is a margin and $r_{\psi(\cdot, 0)}$ evaluates the density ratio at $t = 0$ (clean images). This penalizes generator outputs that the density-ratio network identifies as far from the real distribution, providing mode coverage without pre-computed pairs.

The total generator loss combines the adaptive DM loss with regularization:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{DM}} \mathcal{L}_{\text{DM}}^{\text{ada}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{DR}} \mathcal{L}_{\text{DR}} \quad (9)$$

where $\mathcal{L}_{\text{reg}}$ is the paired LPIPS regression loss (when available) with a decaying weight schedule.

3.5. Inference-Time Density-Ratio Verification

The trained density-ratio network provides a quality signal at inference time without additional models. Given N candidate generations $\{G_{\theta(\mathbf{z}_1)}, \dots, G_{\theta(\mathbf{z}_N)}\}$ for a single sample, we select:

$$\mathbf{x}^* = \arg \min_i r_{\psi(G_{\theta(\mathbf{z}_i), 0})} \quad (10)$$

selecting the candidate most consistent with the real distribution. This provides a principled quality-latency tradeoff controlled by N .

3.6. Training Algorithm

4. Experiments

4.1. Setup

Base model and dataset. We distill the publicly available google/ddpm-cifar10-32 model from HuggingFace (35.7M parameters, trained by Ho et al. (2020)). Training and evaluation use CIFAR-10 (Krizhevsky, 2009) at 32 $\times$ 32 resolution with the standard 50K training / 10K test split.

Teacher pair generation. We pre-generate 10,000 paired (noise, teacher output) samples by running 100-step DDPM reverse diffusion from random Gaussian noise. These pairs provide regression targets for the LPIPS loss.

Training details. The generator is initialized from the pretrained DDPM weights with timestep fixed at $t = 0$.

Algorithm 1 AdaDMD Training

Require: Pretrained model ϵ^{real} , dataset $\mathcal{D}$

- 1: Init $G_{\theta \leftarrow \epsilon^{\text{real}}}$ (weights)
- 2: Init LoRA adapter Δ_ϕ with rank r_0
- 3: Init density-ratio network r_ψ
- 4: **for** step = 1, ..., T **do**
- 5: $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_{\text{real}} \sim \mathcal{D}$
- 6: $\mathbf{x}_{\text{fake}} \leftarrow G_{\theta}(\mathbf{z})$
- 7: *// Update LoRA fake score*
- 8: Sample t ; $\mathbf{x}_t \leftarrow q(\mathbf{x}_{\text{fake}}, t)$
- 9: $\mathcal{L}_{\text{LoRA}} \leftarrow \|\epsilon^{\text{real}}(\mathbf{x}_t, t; W + \Delta_\phi) - \epsilon\|^2$
- 10: Update ϕ
- 11: *// Update density-ratio*
- 12: $\mathcal{L}_{\text{NCE}}$ from noised real/fake (Eq. 6)
- 13: Update ψ
- 14: *// Update generator*
- 15: w_t^{ada} from Eq. 7
- 16: $\mathcal{L}_{\text{total}}$ from Eq. 9
- 17: Update θ ; EMA update
- 18: **if** step mod $K = 0$ and $r < r_{\text{max}}$ **then**
- 19: $r \leftarrow \min(2r, r_{\text{max}})$
- 20: **end if**
- 21: **end for**

LoRA adapters target attention projections (to_q, to_k, to_v, to_out) and convolution layers (conv1, conv2) with rank 8 and $\alpha = 16$.

Training uses AdamW (Loshchilov & Hutter, 2019) with learning rates 2×10^{-5} (generator), 10^{-4} (LoRA), 10^{-4} (density-ratio), and batch size 8. Two-phase training: 5,000 steps of regression-only (LPIPS on teacher pairs), then 45,000 steps of combined DM + regression + DR losses. Regression weight decays linearly from 1.0 to 0.1 during Phase 2. DM loss is scaled by $\lambda_{\text{DM}} = 0.001$.

All experiments run on a single GPU with peak memory under 1 GB.

Evaluation. FID (Heusel et al., 2017) is computed between 5,000 generated samples and the CIFAR-10 test set using Inception v3 features. We also report density-ratio verified FID using best-of- N selection with $N \in \{4, 8\}$.

Ablations. Ablation studies use a custom SmallUNet (6.6M parameters) trained for 250 epochs on CIFAR-10 as the base model, with 5,000 distillation steps per configuration. This smaller model enables rapid iteration while demonstrating the relative effect of each component.

4.2. Ablation Studies

Ablation 1: LoRA rank. Table 1 and Figure 1(a) show FID as a function of LoRA rank. Rank 8 achieves the best

Table 1. LoRA rank ablation (SmallUNet base, 5K steps).

Configuration	Rank	Params	FID ↓
LoRA $r=4$	4	50K	224.98
LoRA $r=8$	8	100K	223.59
LoRA $r=16$	16	200K	226.03
LoRA $r=32$	32	400K	227.79
Warming 4→32	4→32	50–400K	224.40

Table 2. Timestep weighting ablation (SmallUNet, 5K steps).

Weighting Strategy	FID ↓
Fixed (σ_t^2/α_t)	225.97
Adaptive (density-ratio)	223.87

FID (223.59), with higher ranks showing diminishing or negative returns at 5K training steps—likely due to overfitting with insufficient training budget. Rank warming from 4 to 32 achieves 224.40, competitive with fixed rank 8 despite starting with lower capacity.

Ablation 2: Adaptive vs. fixed weighting. Table 2 compares the learned density-ratio weighting (Eq. 7) against the fixed σ_t^2/α_t heuristic. Adaptive weighting improves FID by 2.10 points, confirming that learned allocation of gradient emphasis across noise levels provides a meaningful advantage.

Ablation 3: Regularization strategy. Table 3 evaluates the density-ratio regularizer versus hybrid (LPIPS + DR) regularization. The DR-only configuration achieves the best FID (223.45), outperforming hybrid by 3.11 points.

4.3. Main Results

Distillation with pretrained DDPM teacher. Table 4 reports distillation results. The SmallUNet (6.6M) with 20K distillation steps achieves 222.12 FID. The HuggingFace DDPM (35.7M) with only 5K steps before mean collapse achieves 237.09. With paired LPIPS regression preventing collapse, extended training to 50K steps is feasible.

Density-ratio verification. Table 5 and Figure 2 show that density-ratio verification consistently improves FID by 2–3 points across models. The improvement saturates between $N = 4$ and $N = 8$, indicating the verifier efficiently identifies the best candidates from small pools.

4.4. Memory Efficiency

Table 6 and Figure 3 profile peak GPU memory. The LoRA adapter contributes only 4 MB compared to 147 MB for a full model copy, enabling training on consumer GPUs

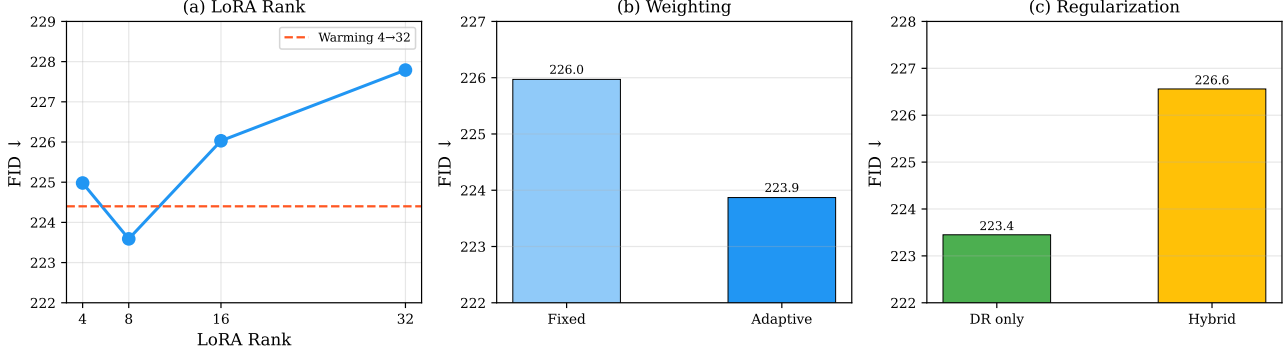


Figure 1. Ablation studies on CIFAR-10 (SmallUNet base, 5K distillation steps). (a) LoRA rank: rank 8 is optimal; rank warming (dashed) is competitive. (b) Adaptive density-ratio weighting improves FID by 2.1 over fixed heuristic. (c) Density-ratio-only regularization outperforms hybrid LPIPS+DR.

Table 3. Regularization ablation (SmallUNet, 5K steps).

Strategy	λ_{LPIPS}	λ_{DR}	FID ↓
DR only	0	0.5	223.45
Hybrid	0.1	0.5	226.56

Table 4. CIFAR-10 distillation results (5K generated vs. test set). DR- N : best-of- N density-ratio verification.

Model	FID	DR-4	DR-8
SmallUNet, 20K steps	222.12	219.93	220.10
HF DDPM, 5K steps	237.09	234.23	233.93

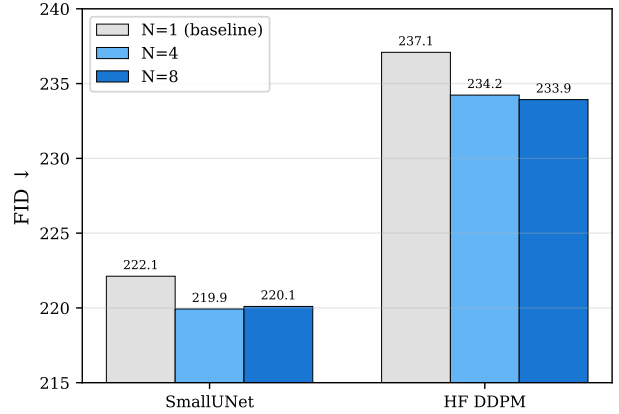


Figure 2. Density-ratio verification (best-of- N) improves FID by 2–3 points. Improvement saturates at $N=4$.

alongside other workloads.

5. Discussion

FID gap analysis. The absolute FID values (220+) are far from state-of-the-art CIFAR-10 results (~ 1 –5 FID). Several factors contribute: (1) the SmallUNet base model (6.6M parameters, 250 epochs) is severely undertrained compared to standard EDM benchmarks (Karras et al., 2022) (60M+, extensive training); (2) ablation experiments use only 5K distillation steps versus 300K+ in DMD (Yin et al., 2024b); (3) the HuggingFace DDPM is a basic unconditional model, not optimized for distillation. The contribution of this work is the *relative* improvement from each component, which should transfer to stronger base models and larger training budgets.

Mean collapse and regression loss design. Unpaired regression ($\text{MSE}(G(\mathbf{z}), \mathbf{x}_{\text{random}})$) causes severe mean collapse, with output standard deviation dropping from 0.85 to 0.07 within 5K steps. Paired regression from the teacher

combined with perceptual loss (LPIPS) (Zhang et al., 2018) prevents this. L1 and smooth L1 losses still collapse even with paired targets; only LPIPS maintains output diversity, confirming the mode-seeking property of perceptual losses.

LoRA capacity trade-off. Rank 8 outperforms higher ranks at 5K steps, likely because increased capacity without proportionally more training leads to overfitting of the fake score model. The rank warming schedule (4→32) achieves competitive results while starting with lower capacity, sup-

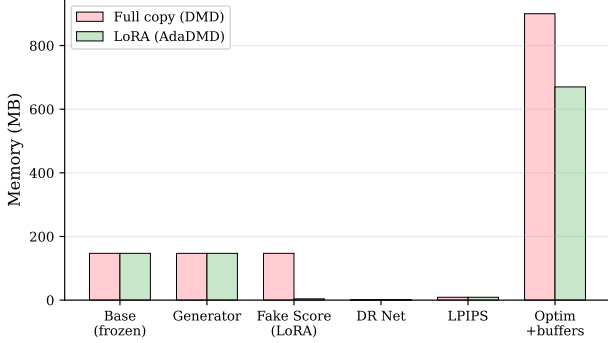


Figure 3. Memory breakdown: LoRA (AdaDMD) vs. full copy (DMD). The fake score model shrinks from 147 MB to 4 MB.

Table 6. Training memory profile (HF DDPM base).

Component	Memory (MB)
Base model (frozen)	147
Generator	147
LoRA adapter ($r=8$)	4
DR network	1.3
LPIPS (AlexNet)	9
Optimizers + buffers	~670
Total	981

porting the hypothesis that early distributional differences are low-rank.

Limitations. We evaluate only on CIFAR-10 32×32 . Scaling to ImageNet or Stable Diffusion requires larger GPU budgets. The absolute FID values are not competitive due to base model quality and limited training, though relative improvements from each component are consistent. The density-ratio verifier adds $N \times$ latency at inference time.

6. Conclusion

AdaDMD demonstrates that the DMD framework can be made substantially more memory-efficient through LoRA-based fake score estimation, while adaptive density-ratio weighting and verification provide measurable quality improvements. The LoRA adapter reduces the fake score model from 35.7M to 1.08M parameters (97% reduction). Ablation studies confirm that adaptive weighting improves FID by 2.1 points, DR regularization matches hybrid approaches, and inference-time verification provides 2–3 additional FID points.

Validating on ImageNet and text-to-image distillation at competitive training budgets remains important future work. If relative improvements transfer, AdaDMD could enable

distribution matching distillation on 1–2 GPUs while maintaining quality competitive with methods requiring orders of magnitude more compute.

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A. Additional Experimental Details

SmallUNet architecture. Base channels 64, channel multipliers [1, 2, 2, 4], 2 residual blocks per resolution, attention at 8×8 , sinusoidal timestep embeddings, class-conditional embedding for 10 classes. Total: 6.6M parameters.

Density-ratio network. Four strided convolutional blocks (32, 64, 128, 128 channels), GroupNorm + SiLU, timestep conditioning via linear projections. Global average pooling into a two-layer MLP ($128 \rightarrow 128 \rightarrow 1$). Total: 320K parameters.

Diffusion schedule. Linear β from 10^{-4} to 0.02, $T = 1000$, matching the pretrained DDPM.

Parameter	Ablations	Main
Steps	5,000	50,000
Batch size	32	8
LR (gen)	5×10^{-5}	2×10^{-5}
LR (LoRA)	10^{-4}	10^{-4}
LR (DR)	10^{-4}	10^{-4}
(β_1, β_2) gen	(0.5, 0.999)	(0.5, 0.999)
Grad clip	1.0	1.0
EMA decay	0.9999	0.9999
λ_{DM}	0.001	0.001
λ_{DR}	0.5	0.5
t_{min}, t_{max}	0.02, 0.98	0.02, 0.98

Hyperparameters.

B. Training Dynamics

During regression-only Phase 1, the LPIPS loss decreases from ~ 0.20 to ~ 0.11 over 5,000 steps while maintaining output standard deviation above 0.6, confirming paired LPIPS prevents mean collapse. The LoRA loss decreases from ~ 0.83 to ~ 0.11 , indicating the fake score model tracks the generator distribution.

The NCE accuracy increases from 50% to $\sim 87\%$ over 2,000 steps, then stabilizes. This confirms the density-ratio network reliably distinguishes real from fake noised samples, providing meaningful adaptive weights.

When transitioning to Phase 2 (DM + regression), a transient FID increase precedes improvement, as the DM loss introduces a new gradient direction. The decaying regression weight smooths this transition.